Prescriptive Modeling: Clustering Cities Using the KMeans Model

Introduction

Clustering cities using the K-Means algorithm is a powerful prescriptive analytics technique that reveals natural groupings based on shared characteristics, such as demographics, purchasing power, and consumer behavior. By uncovering these hidden patterns, organizations can tailor regional strategies, optimize product distribution, and design high-impact interventions aligned with the unique profile of each cluster. This data-driven approach empowers decision-makers to move from one-size-fits-all planning to precise, localized action, significantly boosting operational efficiency and strategic effectiveness.

Let's take a deep dive into the data

Import necessary libraries

In [6]:
```
import pandas as pd
import os
import glob
```

In [7]:
```
folder_path = r"C:\Monthly_Sales"

# Retrieve all CSV files from the folder using glob
all_files = glob.glob(os.path.join(folder_path, "*.csv"))

# All CSV files combined as one DataFrame
all_data = pd.concat([pd.read_csv(file) for file in all_files], ignore_index=True)

# Merged DataFrame saved into a new CSV
output_file = os.path.join(folder_path, "all_data.csv")
all_data.to_csv(output_file, index=False)

print("All files integrated into:", output_file)
```

```
All files integrated into: C:\Monthly_Sales\all_data.csv
```

Load the updated DataFrame

In [9]:
```
# Skip Blank Rows if present in the dataset

df = pd.read_csv(r'C:\Monthly_Sales\all_data.csv', skip_blank_lines=True)
df.head()
```

Out[9]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address
0	175667	iPhone	1	700.0	04/24/24 19:12	135 Meadow St, Boston, MA 02215
1	175668	AA Batteries (4-pack)	1	5.84	04/20/24 13:45	592 4th St, San Francisco, CA 94016
2	175669	AA Batteries (4-pack)	1	5.84	04/28/24 09:17	632 Park St, Dallas, TX 75001
3	175670	AA Batteries (4-pack)	2	5.84	04/23/24 14:06	131 Pine St, San Francisco, CA 94016
4	175671	Samsung Odyssey Monitor	1	409.99	04/23/24 12:13	836 Forest St, Boston, MA 02215

Data Cleaning Process

Thoroughly clean and standardize the data to eliminate errors, ensure consistency, and build a solid foundation for meaningful insights.

Find and remove rows with NaN values

In [12]:
```
df.isna().sum()
```

Out[12]:
```
Order ID            17920
Product Name        17920
Units Purchased     17921
Unit Price          17921
Order Date          17921
Delivery Address    17922
dtype: int64
```

In [13]:
```
# If Nan value is present in Order ID and Unit Purchased, it will be impossible to
# Therefore, drop Nan values in Order ID and Units Purchased.

df.dropna(subset=['Order ID', 'Units Purchased'], inplace=True)

# Check if Nan value is present
df.isna().sum()
```

Out[13]:
```
Order ID            0
Product Name        0
Units Purchased     0
Unit Price          0
Order Date          0
Delivery Address    1
dtype: int64
```

In [14]:
```
# Further check if any NaN values or blank rows are present
```

```
blank_rows_na = df[df.isnull().any(axis=1)]
blank_rows_na
```

Out[14]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address
2195228	Charging Cable	1	14.95	05/24/24 07:04	852 Hickory St, San Francisco, CA 94016	NaN

Find and remove rows with duplicate values

In [16]:
```
# Find duplicate values

df.duplicated()
```

Out[16]:
```
0          False
1          False
2          False
3          False
4          False
           ...
6901236     True
6901237     True
6901238     True
6901239     True
6901240     True
Length: 6883320, dtype: bool
```

In [17]:
```
# Check again for duplicated values

df.drop_duplicates(inplace = True)

# Check again for duplicated values
df.duplicated()
```

Out[17]:
```
0          False
1          False
2          False
3          False
4          False
           ...
172529     False
172530     False
2195228    False
6370082    False
6403570    False
Length: 171545, dtype: bool
```

Verify and fix incorrect data types in the dataset

In [19]:
```
# check for data types
```

```
df.dtypes
```

Out[19]:
```
Order ID            object
Product Name        object
Units Purchased     object
Unit Price          object
Order Date          object
Delivery Address    object
dtype: object
```

Fix incorrect data types

In [21]:
```
df['Order Date'] = pd.to_datetime(df['Order Date'], format='%m/%d/%y %H:%M', errors

df['Units Purchased'] = pd. to_numeric(df['Units Purchased'], errors='coerce')

df['Unit Price'] = pd. to_numeric(df['Unit Price'], errors='coerce')
```

In [22]:
```
# Verify the presence of NaN values remaining in the columns as a result of using e

df.isna().sum()
```

Out[22]:
```
Order ID            0
Product Name        0
Units Purchased     1
Unit Price          2
Order Date          2
Delivery Address    1
dtype: int64
```

In [23]:
```
df = df.dropna()
```

Change the data type to optimize memory usage (Optional)

In [25]:
```
df['Order ID'] = pd.to_numeric(df['Order ID'], downcast='integer')
df['Product Name'] = df['Product Name'].astype('category')
df['Units Purchased'] = df['Units Purchased']. astype('int8')
df['Unit Price'] = pd.to_numeric(df['Unit Price'], downcast='float')
df['Delivery Address'] = df['Delivery Address'].astype('category')
```

Expand the dataset with supplementary columns

In [27]:
```
# Add City column

def city(address):
    return address.split(",")[1].strip(" ")

def state_abbrev(address):
    return address.split(",")[2].split(" ")[1]

df['City'] = df['Delivery Address'].apply(lambda x: f"{city(x)}  ({state_abbrev(x)}
df.head()
```

Out[27]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address	City
0	175667	iPhone	1	700.00000	2024-04-24 19:12:00	135 Meadow St, Boston, MA 02215	Boston (MA)
1	175668	AA Batteries (4-pack)	1	5.84000	2024-04-20 13:45:00	592 4th St, San Francisco, CA 94016	San Francisco (CA)
2	175669	AA Batteries (4-pack)	1	5.84000	2024-04-28 09:17:00	632 Park St, Dallas, TX 75001	Dallas (TX)
3	175670	AA Batteries (4-pack)	2	5.84000	2024-04-23 14:06:00	131 Pine St, San Francisco, CA 94016	San Francisco (CA)
4	175671	Samsung Odyssey Monitor	1	409.98999	2024-04-23 12:13:00	836 Forest St, Boston, MA 02215	Boston (MA)

In [28]:
```
# Add Total Sales column

df['Total Sales'] = df['Units Purchased'] * df['Unit Price']
df.head()
```

Out[28]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address	City	Total Sales
0	175667	iPhone	1	700.00000	2024-04-24 19:12:00	135 Meadow St, Boston, MA 02215	Boston (MA)	700.00000
1	175668	AA Batteries (4-pack)	1	5.84000	2024-04-20 13:45:00	592 4th St, San Francisco, CA 94016	San Francisco (CA)	5.84000
2	175669	AA Batteries (4-pack)	1	5.84000	2024-04-28 09:17:00	632 Park St, Dallas, TX 75001	Dallas (TX)	5.84000
3	175670	AA Batteries (4-pack)	2	5.84000	2024-04-23 14:06:00	131 Pine St, San Francisco, CA 94016	San Francisco (CA)	11.68000
4	175671	Samsung Odyssey Monitor	1	409.98999	2024-04-23 12:13:00	836 Forest St, Boston, MA 02215	Boston (MA)	409.98999

Format Unit Price and Total Sales to 2 decimal places

In [30]:
```
df['Unit Price'] = df['Unit Price'].apply(lambda x: "%.2f" % x)

df['Total Sales'] = df['Total Sales'].apply(lambda x: "%.2f" % x)
df.head()
```

Out[30]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address	City	Total Sales
0	175667	iPhone	1	700.00	2024-04-24 19:12:00	135 Meadow St, Boston, MA 02215	Boston (MA)	700.00
1	175668	AA Batteries (4-pack)	1	5.84	2024-04-20 13:45:00	592 4th St, San Francisco, CA 94016	San Francisco (CA)	5.84
2	175669	AA Batteries (4-pack)	1	5.84	2024-04-28 09:17:00	632 Park St, Dallas, TX 75001	Dallas (TX)	5.84
3	175670	AA Batteries (4-pack)	2	5.84	2024-04-23 14:06:00	131 Pine St, San Francisco, CA 94016	San Francisco (CA)	11.68
4	175671	Samsung Odyssey Monitor	1	409.99	2024-04-23 12:13:00	836 Forest St, Boston, MA 02215	Boston (MA)	409.99

In [31]:
```
# Convert from string to numeric

df['Unit Price'] = pd.to_numeric(df['Unit Price'])
df['Total Sales'] = pd.to_numeric(df['Total Sales'])
```

In [32]:
```
df.dtypes
```

Out[32]:
```
Order ID                    int32
Product Name             category
Units Purchased              int8
Unit Price                float64
Order Date          datetime64[ns]
Delivery Address         category
City                       object
Total Sales               float64
dtype: object
```

Organize Data by Order Date Chronologically and Reindex

In [34]:
```
df = df.sort_values(by = 'Order Date')

df = df.reset_index(drop=True)
df
```

Out[34]:

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address	City	Total Sales
0	160155	Alienware Monitor	1	400.99	2024-01-01 05:04:00	765 Ridge St, Portland, OR 97035	Portland (OR)	400.99
1	151041	AAA Batteries (4-pack)	1	4.99	2024-01-01 05:04:00	964 Lakeview St, Atlanta, GA 30301	Atlanta (GA)	4.99
2	146765	AAA Batteries (4-pack)	1	4.99	2024-01-01 05:20:00	546 10th St, San Francisco, CA 94016	San Francisco (CA)	4.99
3	145617	Amana Washing Machine	1	600.00	2024-01-01 05:24:00	961 Meadow St, Portland, OR 97035	Portland (OR)	600.00
4	156535	iPhone	1	700.00	2024-01-01 05:45:00	451 Elm St, Los Angeles, CA 90001	Los Angeles (CA)	700.00
...	...	...	...	...	...	...	...	...
171538	297748	iPhone	1	700.00	2025-01-01 02:37:00	258 Forest St, Los Angeles, CA 90001	Los Angeles (CA)	700.00
171539	284606	Bose SoundSport Headphones	1	99.99	2025-01-01 02:50:00	211 Johnson St, Boston, MA 02215	Boston (MA)	99.99
171540	302330	AA Batteries (4-pack)	1	5.84	2025-01-01 03:03:00	665 6th St, San Francisco, CA 94016	San Francisco (CA)	5.84
171541	284711	AA Batteries (4-pack)	1	5.84	2025-01-01 03:19:00	250 8th St, San Francisco, CA 94016	San Francisco (CA)	5.84

	Order ID	Product Name	Units Purchased	Unit Price	Order Date	Delivery Address	City	Total Sales
171542	303626	USB-C Charging Cable	3	11.95	2025-01-01 04:43:00	651 Lakeview St, Dallas, TX 75001	Dallas (TX)	35.85

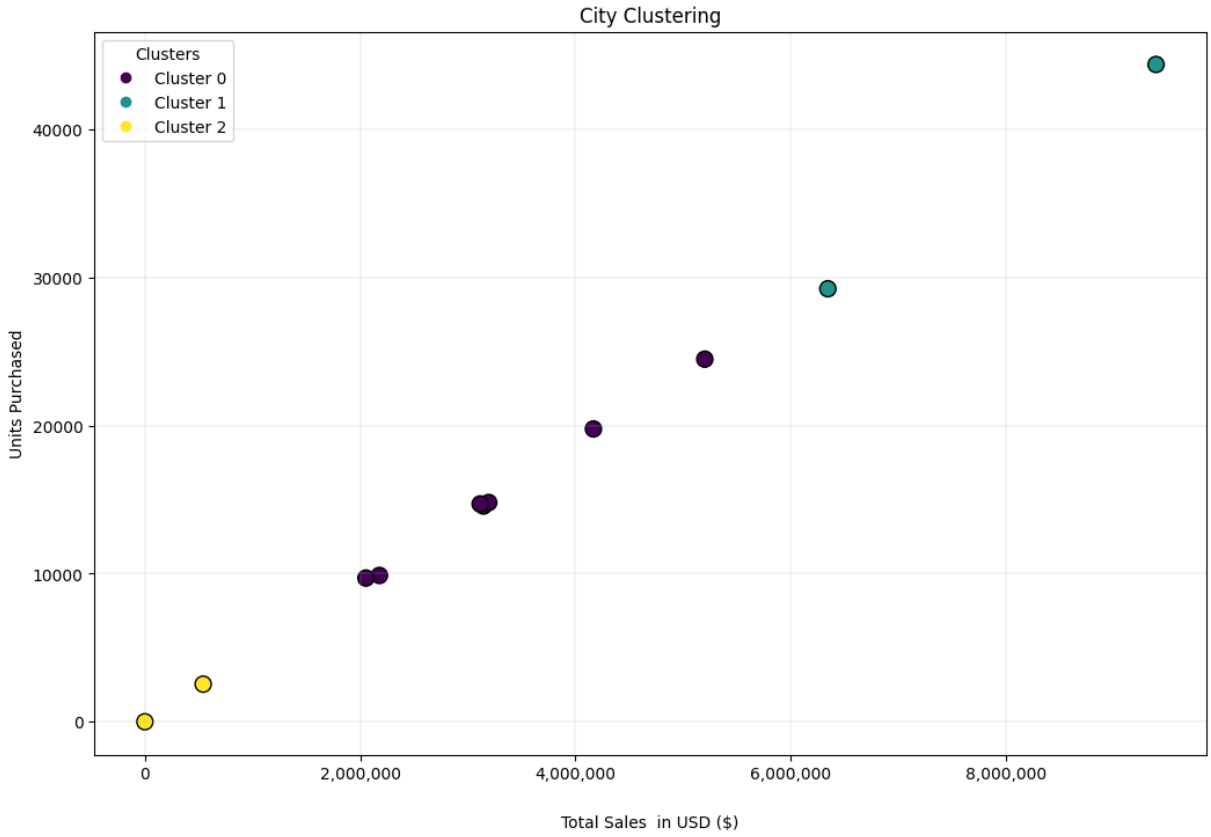
171543 rows × 8 columns

Clustering Cities Using the KMeans Model

QUESTION: How can we group cities based on sales performance to uncover patterns for better sales strategy?

In [59]:
```
import matplotlib.pyplot as plt
from matplotlib.ticker import FuncFormatter
from sklearn.cluster import KMeans

# Deep copy to avoid modifying the original DataFrame
df_cluster = df.copy(deep=True)

# Cluster Cities based on total sales
city_sales_data = df_cluster.groupby('City').agg({'Total Sales': 'sum','Units Purch

city_sales_data.columns = city_sales_data.columns.str.strip()
city_sales_data['City'] = city_sales_data['City'].astype(str).str.replace(r'[\r\n\t

# KMeans clustering
kmeans = KMeans(n_clusters=3, random_state=42)
city_sales_data['Cluster'] = kmeans.fit_predict(city_sales_data[['Total Sales', 'Un

fig, ax = plt.subplots(figsize=(12, 8))

# Scatter plot
df_scatter = ax.scatter(
    city_sales_data['Total Sales'],
    city_sales_data['Units Purchased'],
    c=city_sales_data['Cluster'],
    cmap='viridis',
    s=100,
    edgecolor='black'
)

# Axis labels and formatting
ax.set_xlabel('Total Sales  in USD ($)', labelpad=20)
ax.set_ylabel('Units Purchased')
ax.set_title('City Clustering')
ax.grid(linewidth=0.2)
ax.xaxis.set_major_formatter(FuncFormatter(lambda x, _: f'{x:,.0f}'))

# Legend
```

```
handles, _ = df_scatter.legend_elements()
ax.legend(handles, [f'Cluster {i}' for i in range(len(handles))], title='Clusters')

# Prepare table data
table_data = city_sales_data[['Cluster', 'City', 'Units Purchased', 'Total Sales']]
table_data['Units Purchased'] = table_data['Units Purchased'].apply(lambda x: f'{x:
table_data['Total Sales'] = table_data['Total Sales'].apply(lambda x: f'${x:,.0f}')

cell_text = table_data.values.tolist()
columns = table_data.columns.tolist()

table = plt.table(
    cellText=cell_text,
    colLabels=columns,
    loc='bottom',
    cellLoc='center',
    bbox=[0.0, -0.6, 1, 0.4]  # Lowered the table further
)
table.auto_set_font_size(False)
table.set_fontsize(10)

plt.subplots_adjust(left=0.1, bottom=0.10)

plt.show()
```



Cluster	City	Units Purchased	Total Sales
0	Atlanta (GA)	14,561	$3,145,310
0	Austin (TX)	9,878	$2,178,729
2	Boston (A)	2	$162
0	Boston (MA)	19,767	$4,166,496
0	Dallas (TX)	14,800	$3,193,629
1	Los Angeles (CA)	29,229	$6,343,771
0	New York City (NY)	24,476	$5,200,781
2	Portland (ME)	2,541	$541,532
0	Portland (OR)	9,700	$2,053,174
1	San Francisco (CA)	44,368	$9,391,882
0	Seattle (WA)	14,704	$3,114,139

In [ ]: